



A low-noise readout ASIC with nonlinearity compensated pseudo-differential charge-balanced CV converter for MEMS capacitive accelerometers

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ABSTRACT

This paper presents a low-noise, highly integrated readout ASIC with a pseudo-differential charge-balanced C-V converter (CVC) for MEMS capacitive accelerometers. Assisted by a 4-phase clock scheme, the CVC performs pseudo-differential processing of the single-ended output signal from the sensing element, thereby suppressing the sensor-originated common-mode (CM) interference and bias drift, while enhancing immunity to supply noise and electromagnetic interference (EMI). Based on a theoretical analysis of nonlinearity caused by the parasitic capacitance parallel to the main sensing capacitor bridge, a compensation capacitor array with opposite charging polarity is added to the CVC to cancel the effect of the parasitic capacitance, thereby significantly improving the linearity. A 3rd-order $\Delta\Sigma$ ADC is integrated in the ASIC to convert the CVC's output into digits, while digital temperature compensation is performed with the assistance of an on-chip temperature sensor. Fabricated in a 0.18- μm CMOS process, the ASIC consumes a total power of 3.63 mW. Measurement results show that the system achieves a dynamic range of 95.47 dB across a 200-Hz bandwidth and offers a sensitivity of 80 mV/g, while showing an outstanding nonlinearity of only 1.77% before digital nonlinearity calibration over the input range of ± 30 g, a bias drift of 0.77 mg/ $^{\circ}\text{C}$ before digital temperature calibration in the range of -40°C to 125°C , and a bias instability of 31.57 μg . The figure of merit (FoM) is 305.76 pJ.

1. Introduction

Micro-electromechanical systems (MEMS) capacitive accelerometers are indispensable in a wide range of applications, including inertial navigation, automotive safety systems, aerospace instrumentation, and emerging Internet of Things (IoT) devices. In these scenarios, the accelerometers should exhibit ultra-low noise and high linearity to ensure precise motion detection, while maintaining stable performance across wide temperature ranges. In addition, low power and compact size are also important to enable large-scale, low-cost portable systems [1–4]. These requirements impose severe challenges for the design of the readout circuit, which may limit the achievable noise floor, linearity, and robustness of the system.

Generally, the C-V converters (CVC) in the readout circuits for capacitive MEMS accelerometers can be categorized into closed-loop [1] and open-loop architectures [3–14]. Closed-loop force feedback schemes utilize the sensing element as an electrostatic actuator, generating a counteracting force that nulls the input acceleration, which achieves excellent linearity and wide bandwidth but requires high feedback voltages and complex compensation for stability, resulting in elevated chip area and power consumption [1]. In addition to conventional force-feedback schemes, recent sensor-circuit co-design methods have also explored electrostatic damping feedback and microlever-assisted structures to extend the bandwidth without increasing the sensor resonant frequency [15]. In contrast, the open-loop structure is free from the constraints of the closed-loop

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counterparts, and hence becomes a competitive alternative for cost- and power-sensitive systems [3,13,14,16–18]. Recent open-loop readout works have further improved noise and power efficiency by using high-voltage pulse excitation with active noise cancellation [16] and passive oversampling successive approximation techniques [17]. Among reported open-loop schemes, the conventional ΔC approach suffers from the trade-off between dynamic range and linearity, often showing a heavy nonlinearity exceeding 1% of the full scale (FS) [6,12]. The $\Delta C/C$ (also known as “charge-balanced”) method achieves an inherent linear function of the input acceleration by normalizing the sensed capacitance change (ΔC) to the total capacitance (C), thereby enhancing sensitivity and mitigating drift simultaneously. Owing to these advantages, charge-balanced readouts have become a popular solution for high-performance and low-cost MEMS accelerometer interfaces [3,8,11,13,14].

Although the charge-balanced architecture offers several benefits, two critical limitations remain in most reported implementations. First, since the acquired information from the two-capacitor sensing structure is ultimately converted into a single-ended signal, most readout circuits adopt a single-ended structure [8,11,13], in which the high-impedance sensing nodes are highly susceptible to electromagnetic interference (EMI), supply or ground fluctuations, as well as common-mode (CM) noise from the reference or bias circuitry. The fully differential implementation [3,14] can effectively suppress such interferences but requires a dedicated differential MEMS structure with 4 sensing capacitors, leading to increased cost and poor process compatibility. Second, the inevitable parasitic capacitances—particularly those in parallel with the sensing capacitors—introduce pronounced non-idealities. In high-resolution inertial sensors, fringe and interconnect parasitics are often comparable to the sensing capacitance [2], while the parasitic capacitance in all-silicon MEMS accelerometers (e.g., from the silicon substrate to the electrodes) can reach 850 pF [19], which is much larger than the nominal sensing capacitance of 33–36 pF [19]. Without proper grounding, these parasitic capacitors appear in parallel with the sensing capacitance, shunting the detected signal and significantly degrading the CVC's gain. The parallel parasitic capacitance can further lead to a mismatch in the differential sensing capacitance, resulting in a significant bias error (a 100 fF mismatch in a sensor with a sensitivity of 100 fF/g can generate a bias error of 1 g [20]). More importantly, these parallel parasitic capacitances would cause considerable nonlinearity in the transfer characteristics of the charge-balanced readout structure, which hasn't been thoroughly investigated in prior works. As conventional differential capacitance cancellation methods commonly used for offset trimming [1] are difficult to apply in charge-balanced architectures, the residual nonlinearity can often exceed 1% FS in accelerometers with wide dynamic range.

To address the above challenges, this work presents a readout ASIC with a pseudo-differential charge-balanced CVC architecture that can suppress the nonidealities of single-ended accelerometers while preserving their benefits of process compatibility and low cost. Assisted by a 4-phase clock scheme, the CVC performs pseudo-differential processing of the single-ended output signal from the sensing element, thereby suppressing the sensor-originated CM interference and bias drift, while enhancing immunity to supply disturbance and EMI. To mitigate the nonlinearity and other adverse effects caused by the parallel parasitic capacitances, a compensation capacitor array (CCA) with opposite charging polarity is added in the CVC, which injects compensation charge using reference voltages that are complementary to those applied to the sensing capacitors. Besides the CVC, the readout chain in the presented ASIC further integrates a switched-capacitor (SC) programmable-gain amplifier (PGA), a third-order $\Delta\Sigma$ ADC, an on-chip temperature sensor, and digital circuits realizing bias and gain error correction and other functions. The prototype readout ASIC is fabricated in a 0.18- μm CMOS process with 3.3-V devices, and the overall accelerometer achieves nonlinearity of only 1.77‰ before digital nonlinearity calibration for a ± 30 g input range, and 0.77 mg/ $^{\circ}\text{C}$ bias drift from

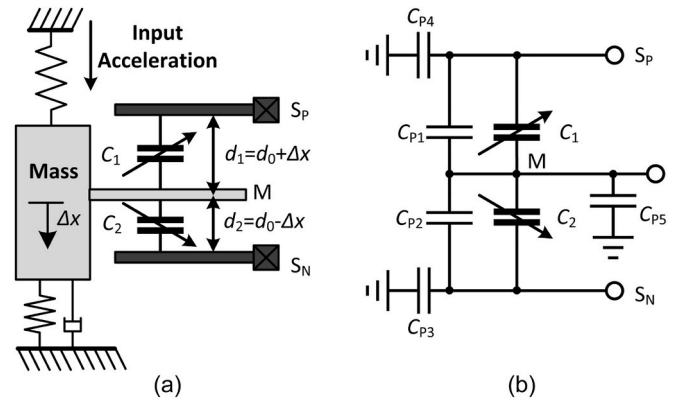


Fig. 1. Principle of MEMS capacitive acceleration sensing: (a) structure; (b) equivalent electrical model.

–40 $^{\circ}\text{C}$ to 125 $^{\circ}\text{C}$, while consuming only 1.1 mA.

This paper is organized as follows. Section II introduces the proposed pseudo-differential charge-balanced CVC with a CCA for nonlinearity compensation. Section III describes circuit details for other important modules in the readout ASIC. Measurement results are presented in Section IV, followed by a conclusion in Section V.

2. Proposed pseudo-differential charge-balanced CVC with nonlinearity compensation

2.1. Analysis of nonideality in charge-balanced CVC

Fig. 1 (a) and (b) illustrate the basic operating principle and

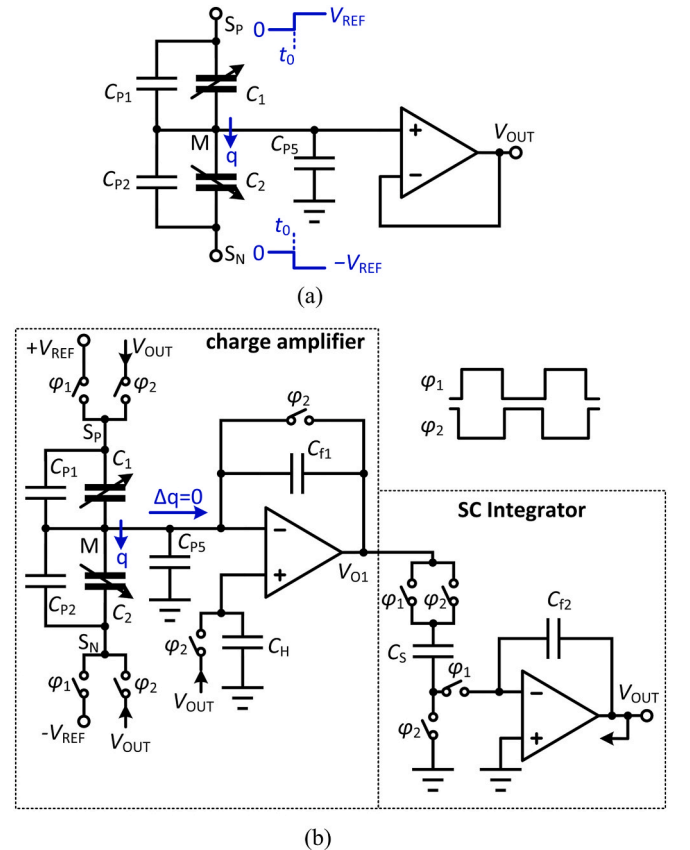


Fig. 2. Charge-balanced CVC: (a) conceptual circuit; (b) the first charge-balanced readout circuit presented in Ref. [11].

equivalent circuit model of a MEMS capacitive accelerometer, respectively. The transducer consists of a moving proof mass suspended by springs above a fixed electrode set. In the quiescent position, the sensor forms 2 equal sensing capacitors with a nominal value of C_S . When subjected to an input acceleration, the proof mass displaces by Δx , resulting in a differential change of ΔC between C_1 and C_2 :

$$\Delta C = C_1 - C_2 = \frac{C_S}{1 + \Delta x/d_0} - \frac{C_S}{1 - \Delta x/d_0} \approx -2 \cdot C_S \cdot \frac{\Delta x}{d_0} \quad (1)$$

where d_0 is the nominal distance between the sensor capacitor electrodes, the conventional ΔC -based CVC can directly translate ΔC into a voltage output [5,6]. However, this approach suffers from inherent nonlinearity. As implied by (1), the error of linear approximation increases with the displacement Δx , leading to increased nonlinearity.

$$(V_{REF} - V_{OUT}[n])(C_1 + C_{P1}) - (V_{REF} + V_{OUT}[n])(C_2 + C_{P2}) = (V_{O1}[n] - V_{OUT}[n]) \cdot C_{f1} \quad (4)$$

Therefore, this method involves a trade-off between linearity and dynamic range. While the force-feedback closed-loop structures [1] can enhance linearity through high loop gain, they require high actuation voltages and complex compensation, which are unsuitable for cost-sensitive applications.

The nonlinearity in the ΔC scheme can be suppressed by the “charge-balanced” method that yields a linear relationship between Δx and $\Delta C/(C_1 + C_2)$:

$$\frac{\Delta C}{C_1 + C_2} = \frac{C_1 - C_2}{C_1 + C_2} = \frac{\frac{C_S}{1 + \Delta x/d_0} - \frac{C_S}{1 - \Delta x/d_0}}{\frac{C_S}{1 + \Delta x/d_0} + \frac{C_S}{1 - \Delta x/d_0}} = -\frac{\Delta x}{2d_0} \propto a \quad (2)$$

Because Δx is proportional to the input acceleration, a , this method provides an inherent linear sensing function of a .

However, those inevitable parasitic capacitors (C_{P1} , C_{P2} , and C_{P5} in Fig. 1(b)) would introduce nonidealities in (2). To perform a detailed analysis of their effect, a conceptual circuit for implementing the charge-balanced CVC is shown in Fig. 2(a), where the voltages at nodes S_P , S_N , and M are initialized to 0 before time t_0 . After the opamp's output settles following complementary reference-voltage switching at S_P and S_N , the output voltage can be derived as

$$V_{OUT} = V_{REF} \cdot \frac{(C_1 + C_{P1}) - (C_2 + C_{P2})}{C_1 + C_2 + C_{P1} + C_{P2} + C_{P5}} + V_{os,opamp} \quad (3)$$

As seen from (3), the idea of the charge-balanced method in (2) can be implemented by the circuit in Fig. 2(a), provided that C_{P1} , C_{P2} , and

embedding the capacitive divider within a negative feedback loop, as illustrated in Fig. 2(b). In this structure, any charge escaping from the divider serves as an error signal [3], which is actively compensated by the closed-loop feedback mechanism, thereby maintaining an accurate and stable charge-balanced state. Its operating principle can be summarized as follows. The first stage is a closed-loop charge amplifier, which routes the escaped error charge onto the input capacitor, C_{in} , of the SC integrator. By exploiting the virtual-short property of the op-amp, C_S is charged to V_{OUT} during phase φ_2 . Therefore, the deviation between V_{OUT} and the desired output $V_{REF}\Delta C/(C_1 + C_2)$ is accumulated by the integrator. When the resulting V_{OUT} equals $V_{REF}\Delta C/(C_1 + C_2)$, the system reaches the charge-balanced state.

Considering C_{P1} and C_{P2} , the charge transfer equations at the N th cycle for the whole CVC during phases φ_1 and φ_2 can be expressed as:

where V_{O1} denotes the output of the closed-loop charge amplifier stage. Meanwhile, the charge transfer equations for the integrator stage during phases φ_1 and φ_2 can be written as:

$$(V_{O1}[n] - V_{OUT}[n])C_{f1} = (V_{OUT}[n+1] - V_{OUT}[n])C_{f2} \quad (5)$$

When the charge-balanced state is achieved, the value of $V_{O1}[n]$ equals $V_{OUT}[n]$. Consequently, according to (4), the single-ended output V_{OUT} becomes

$$V_{OUT} = V_{REF} \frac{(C_1 + C_{P1}) - (C_2 + C_{P2})}{(C_{P1} + C_1) + (C_{P2} + C_2)} \quad (6)$$

Compared to (3), the parasitic capacitor C_{P5} is eliminated, which can be explained as follows. During phase φ_2 , C_{P5} is always charged by V_{OUT} , while its stored charge is preserved since the sensing node is isolated from the sensor capacitor divider during phase φ_1 . As a result, no net charge is injected into or removed from C_{P5} , and the charge-balanced state is not disturbed by C_{P5} . Moreover, the offset error of the opamp, $V_{os,opamp}$ is also eliminated through the auto-zero mechanism ensured by the two-phase clock operating mechanism.

However, the parallel parasitic capacitors C_{P1} and C_{P2} remain in (6), which would affect the ideal linear relationship implied by (2). By substituting $C_1 = C_S/(1 + \Delta x/d_0)$ and $C_2 = C_S/(1 - \Delta x/d_0)$ into (6) and expanding it to a Taylor series around $\Delta x = 0$, the output can be rewritten as:

$$V_{OUT} = V_{REF} \left[\frac{C_{P1} - C_{P2}}{C_T} - \frac{2C_S}{C_T d_0} \Delta x - \frac{2C_S(C_{P1} - C_{P2})}{C_T^2 d_0^2} \Delta x^2 - \left(\frac{4C_S^2}{C_T^2 d_0^3} - \frac{2C_S}{C_T d_0^3} \right) \Delta x^3 + \dots \right] \quad (7)$$

C_{P5} are neglected. Without these parasitic capacitors, V_{OUT} is a linear function of $\Delta C/(C_1 + C_2)$ while the charge q is fully constrained within the capacitive divider. This condition, known as the “charge-balance” state [3], constitutes the ideal operating point that charge-balanced CVC architectures are designed to maintain. However, parasitic capacitors are unavoidable, which introduce severe gain and offset errors, as well as nonlinearity to the V_{OUT} of the conceptual charge-balanced CVC in Fig. 2(a), as implied by (3). Additionally, the offset voltage of the op-amp, $V_{os,opamp}$, also appears in V_{OUT} directly.

The practical charge-balanced CVC architecture in Ref. [11] can mitigate the issues associated with C_{P5} and the opamp-induced offset by

where $C_T = (C_{P1} + C_{P2} + 2C_S)$. As shown, C_{P1} and C_{P2} not only degrade the linear gain ($2C_S < C_T$) but also lead to a nonlinear relationship between V_{OUT} and Δx . To evaluate the nonlinear error, the linear gain (i.e., the coefficient of the linear term) of the CVC can be extracted from (7):

$$K = \frac{2C_S}{C_T} \frac{1}{d_0} V_{REF} \quad (8)$$

The normalized nonlinear error can then be defined as:

$$\varepsilon_{NL} = \frac{|V_{OUT} - \Delta x \cdot K|}{\Delta x_{max} \cdot K} = \frac{C_{P1} - C_{P2}}{2C_S} \frac{d_0}{\Delta x_{max}} - \frac{C_{P1} - C_{P2}}{C_T} \frac{\Delta x^2}{d_0 \Delta x_{max}} - \left(\frac{2C_S}{C_T} - 1 \right) \frac{\Delta x^3}{d_0^2 \Delta x_{max}} + \dots \quad (9)$$

As shown in (9), the nonlinear error terms depend on the values and matching of C_{P1} and C_{P2} . Although the bias shift and even-order nonlinearity can be suppressed by reducing the mismatch ($C_{P1} - C_{P2}$), the third-order component (also other odd-order nonlinear errors) is unavoidable because C_{P1} and C_{P2} can't be zero. The 3rd-order nonlinearity is particularly severe when the parallel parasitic capacitors, C_{P1} and C_{P2} , are comparable to the nominal sensing capacitance [2].

Consequently, although the influence from the grounded parasitic capacitance (C_{P5}) and the opamp offset can be effectively eliminated by the classical charge-balanced CVC in Fig. 2(b), the parallel parasitic capacitances C_{P1} and C_{P2} would still introduce significant nonlinearity. Moreover, since the architecture in Fig. 2(b) produces a single-ended output, it is inherently susceptible to the EMI, supply and ground noise, as well as disturbances from V_{CM} or reference voltages. Recent works with charge-balanced CVC have addressed some of these limitations. For instance, the fully differential implementations in Refs. [3,14] improve robustness to EMI by employing differential-sensing structures, whereas [13] adopts a random-chopping technique to suppress EMI using conventional sensors. However, these approaches mainly focus on

improving EMI immunity and do not explicitly address or compensate for the nonlinearity introduced by the parasitic capacitances.

2.2. Proposed pseudo-differential charge-balanced CVC with nonlinearity compensation

To suppress low-frequency CM and supply disturbances, this work presents a pseudo-differential charge-balanced CVC structure based on a 4-phase clock scheme for the 2-capacitor accelerometer. The circuit structure is shown in Fig. 3, which employs 3 OTAs to convert the single-ended signal at the sensing node M into a differential output. By interleaving 4 non-overlapping clock phases ($\phi_1 \sim \phi_4$), the scheme excites the 2 sensing capacitors, C_1 and C_2 , by reference voltages $+V_{REF}$ ($V_{REF} - GND$) and $-V_{REF}$ ($GND - V_{REF}$) alternatively in a clock period, generating a differential output voltage of $V_P - V_N$ after ϕ_4 . As a result, the low-frequency CMI and supply noise can be suppressed effectively without incurring complex four-capacitor MEMS accelerometers. In order to eliminate the influence from the offset voltages of OTA₁ and OTA₂, autozero operations are performed using the 2 capacitors, C_{Z1} , which store the offset voltages of OTA₁ and OTA₂ in ϕ_{1N} and ϕ_{3N} , respectively. The offset voltage and low-frequency noise of OTA₃ are suppressed by the chopper switches operating at a frequency that is half of the main clock.

Unlike the 4-phase clock scheme in the touch sensor readout circuit

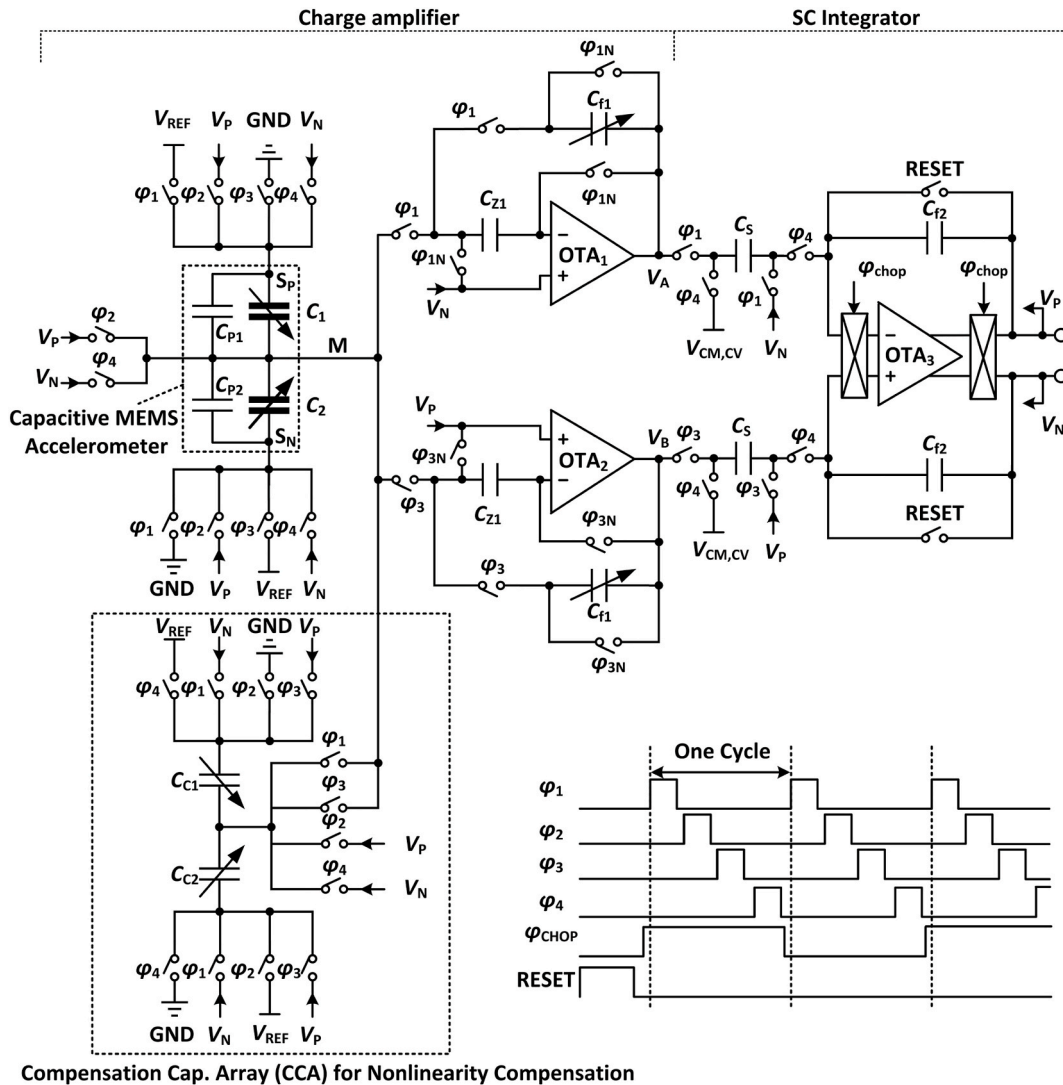


Fig. 3. Circuit and timing diagram of proposed pseudo-differential four-phase charge-balanced CVC with nonlinearity compensation.

in Ref. [21], the proposed circuit is derived from the charge-balanced CVC in Fig. 2(b), in which the output voltages V_P and V_N are fed back to several internal nodes of the CVC to realize the charge-balanced operation. As a result, the grounded parasitic capacitance at node M does not affect the CVC's output. The four phases are organized into 2 complementary charge-balance operations. The first operation is carried out in φ_1 and φ_2 phases, while the second operation is carried out in φ_3 and φ_4 phases. The combinations of reference voltages and output voltages for the 2 sensing capacitors have opposite polarities in the 2 operations.

To suppress the nonlinearity caused by the parallel parasitic capacitors, C_{P1} and C_{P2} , a CCA is added into the CVC (inside the dash-box in Fig. 3), which generates compensation charge using reference voltage polarities that are complementary to those applied to the main sensing capacitors.

To demonstrate the theoretical effectiveness of the proposed CVC, the differential output expression is derived as follows. Similar to (4), the pseudo-differential voltage at the first stage's output (V_A and V_B) at the N th cycle can be derived as

$$(V_A[n] - V_N[n])C_{f1} = (V_{REF} - V_N[n])(C_1 + C_{P1}) - V_N[n](C_2 + C_{P2}) - (V_{REF} - V_N[n])C_{C1} + V_N[n]C_{C2} \quad (10)$$

$$(V_B[n] - V_P[n])C_{f1} = (V_{REF} - V_P[n])(C_2 + C_{P2}) - V_P[n](C_1 + C_{P1}) - V_P[n]C_{C1} + (V_{REF} - V_P[n])C_{C2} \quad (11)$$

where C_{f1} denotes the feedback capacitor of OTA₁ and OTA₂. The input capacitors, C_S , of the SC integrator in the positive and negative paths are charged by $V_A[n]-V_N[n]$ and $V_B[n]-V_P[n]$ in φ_1 and φ_3 phases, respectively. Then in the φ_1 phase, the error charge associated with the desired output $V_{OUT}(=V_P-V_N)$ is accumulated onto C_{f2} , and can be expressed as:

$$(V_A[n] - V_N[n])C_{in} = (V_P[n+1] - V_P[n])C_{f2} \quad (12)$$

$$(V_B[n] - V_P[n])C_{in} = (V_N[n+1] - V_N[n])C_{f2} \quad (13)$$

When the structure reaches the charge-balanced state, no charge accumulates in C_{f2} , so $V_P-V_N = V_A-V_B$ after the system settles. Therefore, according to (12) and (13), the sensed output of the CVC can be derived as

$$V_{OUT} = V_{REF} \frac{(C_1 + C_{P1} - C_{C1}) - (C_2 + C_{P2} - C_{C2})}{(C_1 + C_{P1} - C_{C1}) + (C_2 + C_{P2} - C_{C2})} \quad (14)$$

As shown, the CCA (including C_{C1} and C_{C2}) provides a means to cancel the parasitic capacitors (C_{P1} and C_{P2}) parallel to the main sensing capacitors (C_1 and C_2).

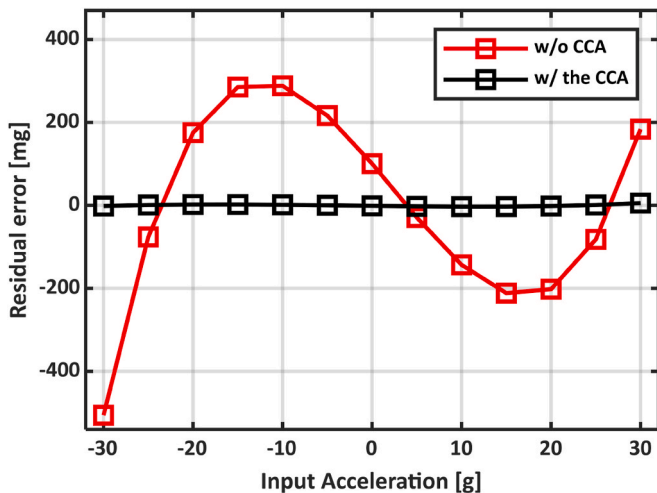


Fig. 4. Simulated residual error of the readout results using a commercial MEMS accelerometer sensor with and without the CCA.

The calibration procedure is performed during factory trimming to determine the tuning codes for the CCA. With the CCA disabled, the MEMS sensor is tested under ± 1 g, and 0 g inputs. The outputs at ± 1 g are used to extract the gain error, while the 0 g output is used to determine the offset error. These errors are then used to estimate parasitic capacitances C_{P1} and C_{P2} , since they jointly determine the converter gain and offset errors. Based on the estimated C_{P1} and C_{P2} , the tuning codes for compensation capacitors C_{C1} and C_{C2} are calculated and programmed into the CCA to enable nonlinearity compensation. Since this procedure relies only on externally measurable gain and offset errors, it can be executed automatically for each packaged sensor during batch factory calibration without directly measuring the parasitic capacitances.

Fig. 4 presents the simulated residual error for a commercial MEMS sensor with an input range of -30 g to 30 g, where the nominal sensing capacitors C_1 and C_2 are 12 pF, and the parasitic capacitances C_{P1} and C_{P2} are approximately 16 pF and 18 pF, respectively. Without calibration, the maximum nonlinear error reaches 1.68% (505.9 mg). After enabling the CCA using the estimated tuning codes obtained during factory calibration, the nonlinear error is reduced to 0.02% (5.45 mg). The remaining error is mainly limited by the finite resolution of the compensation array (10-bit with an LSB of 32 fF in this work). In addition to reducing the nonlinearity, the calibration also improves the sensitivity from 12.5 mV/g to 26 mV/g. Therefore, the proposed CCA suppresses most of the large nonlinear distortion in the analog domain, effectively compressing the residual nonlinearity range and thereby relaxing the calibration range required for the subsequent digital nonlinearity correction.

2.3. Bandwidth and stability of the proposed CVC

After correct compensation, the effects of the parallel parasitic capacitance can be neglected. Combining (13) and (14), the state equation for the pseudo-differential CVC can be rewritten as:

$$V_{OUT}[n+1] - V_{OUT}[n] = [(V_A[n] - V_B[n]) - V_{OUT}[n]] \cdot \frac{C_S}{C_{f2}} = V_{REF} \frac{(C_1 - C_2)C_S}{C_{f2}C_{f1}} - V_{OUT}[n] \frac{(C_1 + C_2)C_S}{C_{f2}C_{f1}} \quad (15)$$

The z-transform $A_{CV}(z)$ of the proposed CVC can be derived from (15):

$$A_{CV}(z) = \frac{V_{OUT}}{-\Delta x/d_0}(z) = V_{REF} \frac{A}{z - (1 - A)} \quad (16)$$

where $A = (C_1 + C_2)C_S / (C_{f2}C_{f1})$. Then, the -3-dB bandwidth, f_c , of the CVC can be derived as

$$f_c = \frac{1}{2\pi} \frac{A}{1 - A} f_s \quad (17)$$

where f_s denotes the effective sampling frequency of the CVC, which is one quarter of the master clock frequency due to the four non-overlapping phases derived from the master clock. The system is stable if the pole of its transfer function lies within the unit circle. This condition determines the maximum value of the capacitance ratio, which can be obtained from (16).

$$A = \frac{(C_1 + C_2)C_{in}}{C_{f2}C_{f1}} < 2 \quad (18)$$

To accommodate sensors with varying capacitance values and resonant frequencies, C_{f1} in this design is configurable, while the integrator gain coefficient, C_{in}/C_{f2} , is set to $1/4$. This setup provides a suitable margin for system stability in practical applications.

2.4. Noise analysis of proposed CVC

The noise of the proposed pseudo-differential four-phase charge-balanced CVC originates mainly from 4 major contributors: the reference block, the sensor-side sampling network, the closed-loop charge amplifier, and the integrator. Each module further generates both amplifier noise and noise induced by the sampling switches. Denoting the equivalent input-referred noise power of these four modules as N_{REF} , N_{SW} , N_{CA} , and N_{INT} , respectively, the total output-referred noise power of the CVC, $N_{CV,OUT}$, can be expressed as:

$$N_{CV,OUT} = \frac{(N_{REF} + N_{SW} + N_{CA}) \cdot G_{CA}^2 \cdot G_{INT}^2}{1 - z^{-2} \cdot G_{INT}^2 - z^{-2} \cdot G_{CA}^2 \cdot G_{INT}^2} + \frac{N_{INT} \cdot G_{INT}^2}{1 - z^{-2} \cdot G_{INT}^2 - z^{-2} \cdot G_{CA}^2 \cdot G_{INT}^2} \quad (19)$$

where G_{CA} denotes the gain of the closed-loop charge amplifier, and G_{INT} represents the gain coefficient of the integrator stage. These two gains can be expressed as $G_{CA} = 2 \cdot C_S / C_{F1}$ and $G_{INT} = (C_{F2} + C_{in}) / C_{F2}$, respectively.

Typically, the dominant noise contributors are the sampling switches at the sensor interface, the reference-path noise, and the first-stage closed-loop charge amplifier. In contrast, the noise of the second-stage integrator is largely suppressed by the gain of the preceding stages and the overall loop gain. In the proposed design, the sampling-switch noise is reduced by the sensor, which has a relatively large sensing capacitance and its associated parasitics. The low-frequency noise in the reference path is mitigated through correlated sampling enabled by the two complementary charge-balance conversion phases. The contribution of $1/f$ noise from the 3 OTAs is further suppressed by the auto-zeroing operation in the first-stage charge amplifier and the chopping technique applied in the second stage. In addition, the pseudo-differential time-division structure inherently cancels low-frequency V_{CM} -related noise. As a result, the critical low-frequency noise components, including $1/f$ noise, are substantially attenuated, and the overall system noise is dominated by thermal noise. This analysis of noise is consistent with both simulation and measurement results, confirming the effectiveness of the proposed architecture.

These noise characteristics also translate into specific design requirements for the analog front end. The OTA₁ and OTA₂ must provide high open-loop gain and a wide output swing to minimize the influence of C_{P5} and to accommodate the possible large swings of V_A and V_B . Therefore, OTA₁ and OTA₂ are implemented using a conventional two-stage architecture with Miller compensation, shown in Fig. 5 (a). The input stage uses a folded-cascode topology to provide high intrinsic gain and a wide input range, while the second stage employs a class-AB output stage to enhance the open-loop gain further and achieve rail-

to-rail output swing, accommodating large swings in V_A and V_B . The design requirement for OTA₃ is driven not by noise but by the need to settle with high accuracy while driving the MEMS sensor's capacitive load. Therefore, OTA₃ is designed as a two-stage fully differential amplifier with a class-AB output stage, similar to OTA₁ and OTA₂. An integrated common-mode feedback circuit stabilizes the common-mode voltage, while the fully differential configuration improves common-mode noise rejection. The complete circuit implementation is shown in Fig. 5(b). The OTAs are designed to achieve an open-loop gain above 100 dB and a unity-gain bandwidth exceeding 10 MHz to ensure sufficient settling accuracy.

The reference voltages $V_{CM,CV}$, and V_{CM} are generated by on-chip reference buffers to ensure stable biasing of the analog front end and to suppress the influence of supply variations. Owing to the pseudo-differential charge-balance method, the V_{CM} is set to half of V_{REF} . The complete CVC operates at 500 kHz, consumes 380 μ A, and achieves a simulated noise density of 200 nV/ $\sqrt{\text{Hz}}$ with 30-pF sensor capacitance.

3. ASIC architecture and circuit design

3.1. System overview of the proposed accelerometer

The overall architecture of the readout ASIC, including the proposed pseudo-differential charge-balanced CVC with nonlinearity compensation, is illustrated in Fig. 6. The CVC in the analog frontend serves as the primary transducer, converting mechanical acceleration into an electrical signal. A programmable-gain amplifier (PGA) is implemented after the CVC to relax the output swing requirements of the CVC, suppress the noise contribution of the subsequent ADC, and enable fine adjustment of sensitivity. A 3rd-order single-bit $\Delta\Sigma$ modulator ($\Delta\Sigma$) then performs high-resolution analog-to-digital conversion of the PGA's output. The analog section also integrates several essential on-chip supporting blocks, including an oscillator (OSC), a reference generator, and several LDO regulators. The reference generator produces a 2.5-V reference used by both the capacitance-to-voltage (CV) converter and the $\Delta\Sigma$ modulator. At the same time, the LDO regulators provide stable supply voltages and generate the required common-mode voltage for the CVC and other blocks.

To support real-time compensation for the sensor's temperature dependence, a compact, low-cost on-chip temperature sensor is designed. In the digital domain, dedicated filters are implemented to process the outputs of both the $\Delta\Sigma$ and the temperature sensor. A temperature-calibration engine compensates for the accelerometer's sensitivity and bias drift across temperature variations. In addition, a nonlinearity-correction logic block is integrated to remove the residual nonlinearity of the readout interface. Finally, an SPI interface enables reliable communication between the ASIC and the host system. The

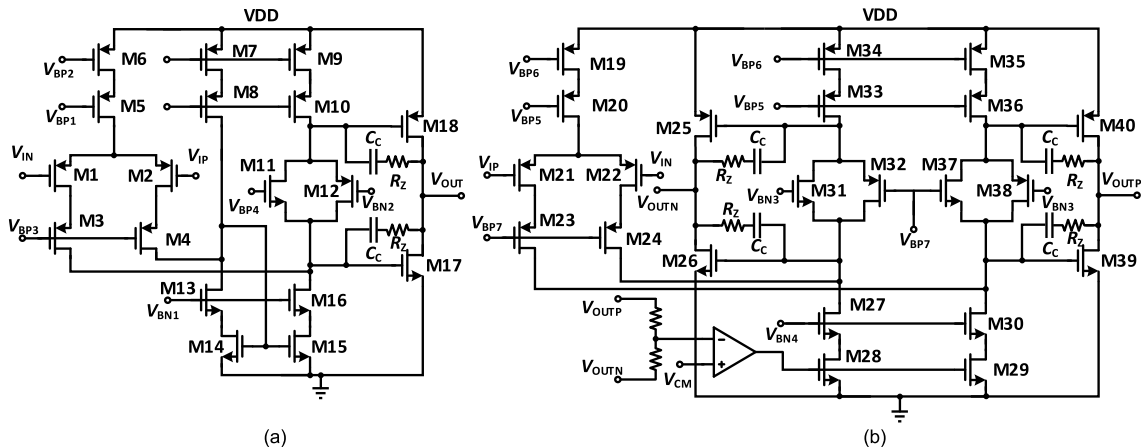


Fig. 5. Structure of the OTAs employed in the proposed CVC: (a) Architecture of OTA1 and OTA2; (b) Architecture of OTA3.

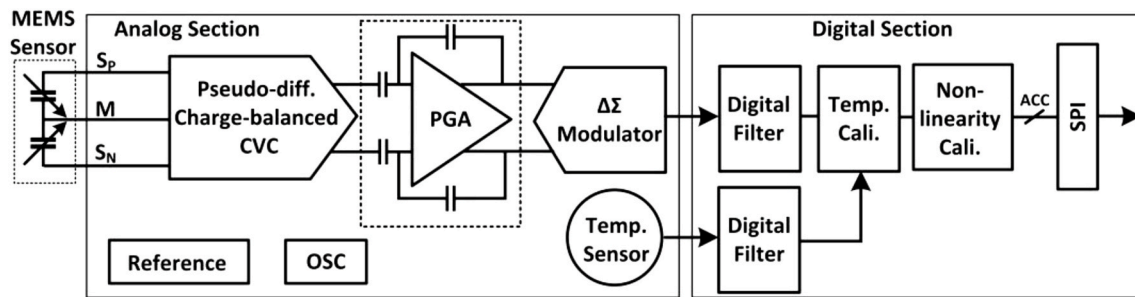


Fig. 6. System architecture of the presented readout ASIC for MEMS capacitive accelerometers.

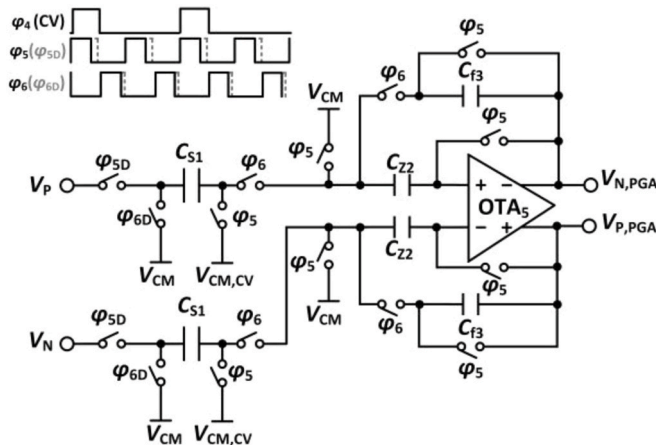


Fig. 7. Circuit of the SC PGA with timing diagram.

detailed architectures of the main building blocks are described in the following subsections.

3.2. Programmable-gain amplifier

Since the intrinsic gain of the CVC is fundamentally limited by the sensor capacitance, additional amplification is required to ensure adequate signal amplitude for the following $\Delta\Sigma\text{M}$. In this work, a discrete-time switched-capacitor PGA is adopted, shown in Fig. 7, following an architecture similar to that presented in Ref. [22]. The PGA samples the CVC's output during phase ϕ_{S} and, unlike conventional SC

PGAs, incorporates an embedded auto-zero operation during the reset phase φ_6 . This nested auto-zeroing effectively suppresses the OTA's offset and $1/f$ noise, thereby enhancing overall accuracy and noise performance. The gain of this PGA can be approximately derived as $-C_{S1}/C_{F3}$.

With the OTA₄ in the PGA designed as a two-stage architecture to accommodate a large output swing, the PGA consumes 160-μA power under 1-MHz operation frequency, achieving a simulated output noise density of 400 nV/√Hz with a nominal gain of 3.2. To enable fine-tuning of sensitivity, the PGA gain is adjustable from 1 to 10 with a fine step of 0.01 by adjusting the binary-weighted capacitor C_{S1}. Although the CVC operates at 500 kHz, its four-phase operation provides two settled output windows at phases φ_2 and φ_4 . The SC PGA and the following ΔΣM operate at 1 MHz with sampling instants synchronized to these windows, ensuring that the settled CVC output is sampled and aligned with the main clock signal.

3.3. The third-order $\Delta\Sigma$ modulator

After being amplified by the SC PGA, the transducer's output is digitized by the 3rd-order $\Delta\Sigma$ with a cascade-of-integrators feedforward (CIFF) structure, as shown in Fig. 8. The 1-bit feedback digital-to-analog converter (DAC) shares the reference voltage with the CVC, ensuring a stable ratio-metric signal transfer function against variations in the DC reference voltages. Additionally, the ratio C_{S2} over C_{S3} provides a passive gain for the ADC, functioning as a secondary passive gain amplifier (PGA) to realize fine-tuning sensitivity. This design mitigates possible drift or noise from the reference voltage, preventing it from affecting the output. Furthermore, the ADC's first integrator employs a chopper structure to suppress offset and $1/f$ noise. The complete ADC operates at 1 MHz to fulfill the system's quantization noise requirement.

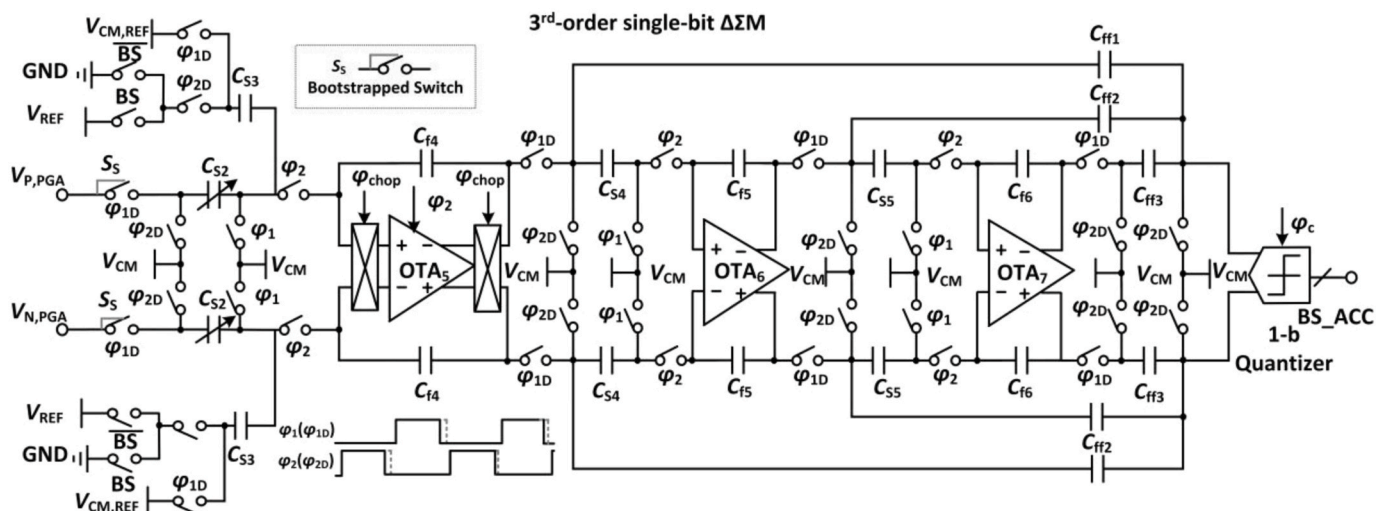


Fig. 8. Circuit structure of the 3rd-order $\Delta\Sigma$ M in the main signal chain.

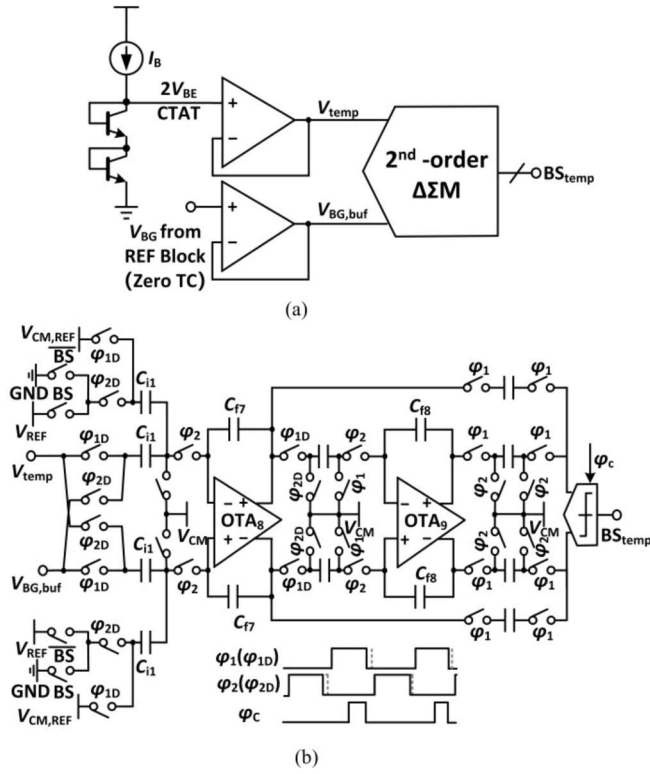


Fig. 9. Analog circuit of the temperature sensor: (a) overall structure; (b) $\Delta\Sigma$ modulator used in the temperature sensor.

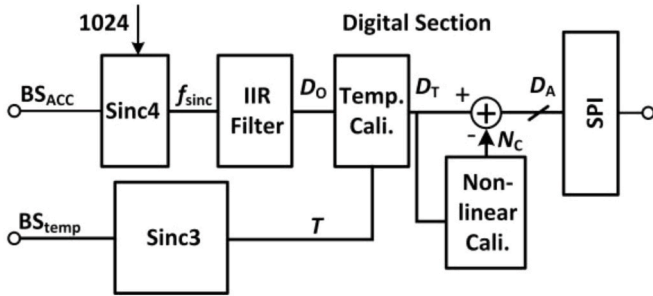


Fig. 10. Diagram of the main digital circuits.

It utilizes a 1.8-pF sampling capacitor and consumes 390 μ A, while achieving a simulated SNR of 118 dB with an OSR of 2500.

3.4. Compact temperature sensor

To compensate for the overall temperature drift of the accelerometer, a compact temperature sensor is integrated on-chip, as shown in Fig. 9. The temperature sensing frontend employs two stacked BJTs with a bias current source, ensuring that the buffered output, V_{temp} , is a complementary-to-absolute-temperature (CTAT) voltage. A bandgap voltage with low temperature coefficient (TC), V_{BG} , from the reference block is adopted as the reference voltage of the $\Delta\Sigma$. The difference between V_{temp} and the buffered bandgap voltage, $V_{BG,buf}$, is digitized directly using a second-order $\Delta\Sigma$ modulator with double sampling [23]. This method eliminates the need for an additional CM voltage reference. Since temperature signals operate at very low frequencies, a 2nd-order $\Delta\Sigma$ is enough to achieve the required accuracy with a high oversampling ratio (OSR) and a double sampling technique. The TC of V_{BG} can be tolerated because the temperature sensor's output is used solely to calibrate the accelerometer's temperature variations rather than to

measure ambient temperature. The temperature sensor operates at an output data rate (ODR) of 30.5 Hz with a current consumption of only 30 μ A, achieving an RMS noise of 0.03 $^{\circ}$ C.

3.5. Digital circuits including temperature calibration

Fig. 10 illustrates the structure of the digital section, which consists of a Sinc filter and an infinite impulse response (IIR) filter for processing the main $\Delta\Sigma$'s output. The bit stream from the $\Delta\Sigma$, BS_{ACC} , is first decimated to reduce the data rate. The digital signals undergo further decimation to reach the final ODR. To streamline the digital circuitry, the temperature sensor's bit stream is handled using a sinc3 digital filter [24]. The temperature calibration process within the digital section is performed according to the following expression:

$$D_T = (D_O - B_C(T)) \cdot \left(1 + \frac{G_C(T)}{G_O}\right) \quad (20)$$

where D_O represents the accelerometer's original digital output, D_T is the digital output after calibration, T is the temperature sensor's digital output, and G_O is the desired sensitivity, G_C and B_C correspond to the third-order polynomial fittings of the accelerometer's gain drift and bias drift. Obviously, G_C and B_C are functions of temperature. The intrinsic temperature dependence of the overall accelerometer can be obtained by recording the D_O values across the full measurement range while using a dividing head. The gain drift, $G_C(T)$, is modeled as a third-order polynomial.

$$G_C(T) = G_0 + G_1 \cdot T + G_2 \cdot T^2 + G_3 \cdot T^3 \quad (21)$$

where G_0 , G_1 , G_2 , and G_3 represent the 0th, 1st, 2nd, and 3rd-order coefficients of the 3rd-order polynomial. Similarly, the bias drift is also modeled as a 3rd-order polynomial as.

$$B_C(T) = B_0 + B_1 \cdot T + B_2 \cdot T^2 + B_3 \cdot T^3 \quad (22)$$

where B_0 , B_1 , B_2 , and B_3 are the coefficients of the 3rd-order polynomial.

The nonlinear calibration logic within the digital section operates as follows:

$$D_A = D_T - N_C(D_T) \quad (23)$$

where N_C is the compensation value of the nonlinearity error. The accelerometer's inherent nonlinear relationship can be determined by recording its outputs in response to specified input accelerations across

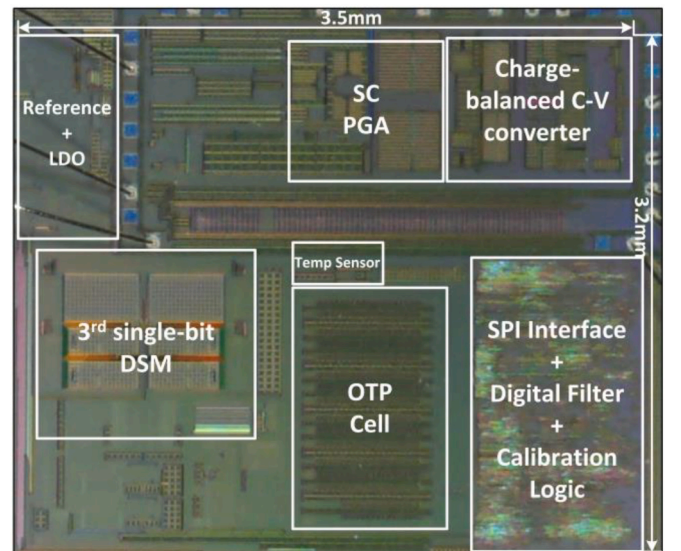


Fig. 11. Photography of the proposed ASIC.

the full measurement range [25]. By applying the least-squares fitting method, the nonlinearity correction value N_C can be modeled as a third-order polynomial in the measured acceleration.

$$N_C(D_T) = N_0 + N_1 \cdot D_T + N_2 \cdot D_T^2 + N_3 \cdot D_T^3 \quad (24)$$

where N_0 , N_1 , N_2 , and N_3 represent the coefficients of the 3rd-order polynomial, corresponding to the 0th, 1st, 2nd, and 3rd orders, respectively. All coefficients are stored in the one-time programmable (OTP) cells in the test phase. Since the proposed CCA already suppresses the dominant nonlinear components at the analog front end, the residual nonlinearity to be corrected in the digital domain becomes significantly smaller. This allows the digital calibration to operate within a much narrower correction range, enabling more accurate compensation with limited hardware complexity.

4. Measurement results

The readout ASIC for MEMS capacitive accelerometers is fabricated using a 0.18- μm CMOS process, incorporating 3.3 V MOS devices. As illustrated by the die micrograph in Fig. 11, the core interface circuitry, including power management and OTP blocks, occupies an area of 8.32 mm² and consumes 1.1 mA from a 3.3-V supply. The pseudo-differential charge-balanced CVC accounts for 34% of the total power consumption, followed by the PGA at 15%, the digital block at 7%, and the $\Delta\Sigma\text{M}$ at 35%. The remaining power is distributed among the reference generation and power management circuits.

The proposed readout ASIC is co-packaged with a commercial MEMS capacitive accelerometer, and all the following measurement results therefore represent system-level performance under realistic operating conditions. Fig. 12 plots the measured output noise spectrum of the overall accelerometer, obtained by recording the output under a 0-g input condition for 100 s at a 1-kHz ODR with a 200-Hz bandwidth digital filter. The measured noise floor is 28.5 $\mu\text{g}/\sqrt{\text{Hz}}$, where the CVC and the PGA primarily dominate. The desired signal has a full-scale range of ± 30 g, providing a dynamic range of 95.47 dB over a 200 Hz bandwidth. Fig. 13 shows the measured bias instability of the overall accelerometer. The curve in Fig. 13(a) is derived from Allan Deviation analysis of the measured 4-h output with zero acceleration at room temperature. The accelerometer system's bias instability is approximately 31.57 μg at around 10 s. The 1-s smoothed result has a 1- σ value of approximately 47.5 μg , as shown in Fig. 13(b).

Fig. 14(a) shows the measured output of the overall accelerometer with different conditions, verified using a centrifuge over a ± 30 g input acceleration range, along with a linearly fitted curve. Fig. 14(b) presents

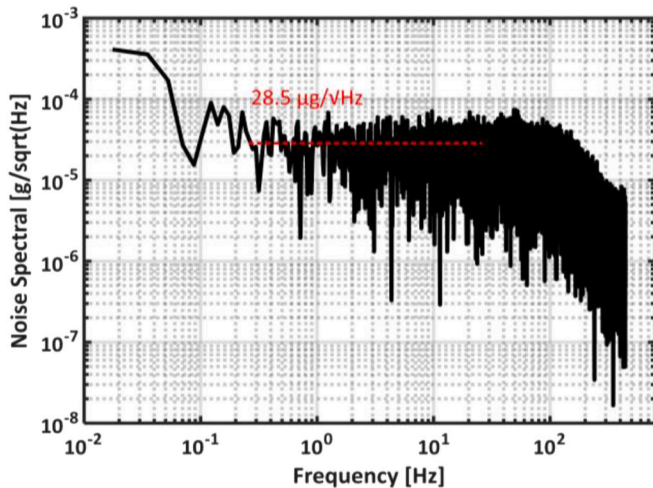


Fig. 12. Measured Output noise spectrum of the proposed accelerometer at 1 kHz ODR with a 200-Hz bandwidth digital filter.

the corresponding residual error. The dominant nonlinearity arises from the parallel parasitic capacitances coupled to the sensing capacitors, resulting in a nonlinearity of 1.77% over the ± 30 g range without compensation. To suppress this parasitic-induced nonlinearity at the circuit level, the CCA in Fig. 3 can be enabled to show its effect. With the CCA enabled, the inherent nonlinearity is substantially reduced to 1.77%. By removing most of the large nonlinear distortion in the analog domain, the proposed CCA effectively compresses the residual nonlinear range, significantly reducing the correction range required for subsequent digital calibration. A subsequent digital nonlinearity calibration further compensates residual mismatches and higher-order effects, reducing the measured nonlinearity to an excellent value of 0.089% over the ± 30 g input range. It should be noted that the measured residual error in Fig. 14(b) is slightly larger than the simulated value of 5.45 mg shown in Fig. 4. This is mainly due to the simulation result reflecting only the effect of the parasitic-capacitance-induced nonlinearity and the finite resolution of the CCA under circuit-level conditions. In contrast, the measured result is obtained from the complete accelerometer system and therefore includes additional practical nonidealities, such as MEMS mechanical nonlinearity, sensing-structure asymmetry, package-induced stress, CCA capacitor mismatch, and residual parasitic-parameter estimation error.

Fig. 15 presents the bias drift measurements for 9 accelerometer chips across a temperature range of -40 to 125 $^{\circ}\text{C}$, showing a minimum bias temperature drift of 0.77 mg/ $^{\circ}\text{C}$ before digital temperature calibration in (a) and 0.006 mg/ $^{\circ}\text{C}$ after digital temperature calibration in (b). The nine tested chips were fabricated in the same batch and characterized using the same measurement setup. The temperature calibration was performed using nine reference temperature points within the range of -40 $^{\circ}\text{C}$ to 125 $^{\circ}\text{C}$, and all measured chips adopted the same third-order temperature-drift fitting with chip-specific coefficients extracted individually. The sample-to-sample bias-drift variation is mainly attributed to MEMS fabrication variations and thermo-mechanical differences induced in the packaging and assembly process, while the residual drift after calibration may arise from finite-order fitting, limited on-chip calibration resources, and possible transient temperature differences between the ASIC-integrated temperature sensor and the MEMS sensing element during rapid or large temperature variations.

The overall performance of the proposed accelerometer is summarized in Table I with comparison to some representative state-of-the-art designs. By combining the charge-balanced method with the proposed CCA for nonlinearity compensation, the accelerometer achieves an inherent nonlinearity of 1.77% over a ± 30 g input range before digital calibration, demonstrating superior linearity compared with recently reported open-loop structures and the conventional ΔC -based method in Ref. [12]. Among reported charge-balanced designs [3,9,13], this result achieves one of the best linearity performances without digital calibration, highlighting the effectiveness of the proposed circuit-level compensation. With an additional lightweight digital calibration, the nonlinearity is further reduced to 0.089%, achieving the best linearity performance among the compared designs. In terms of noise performance, this work achieves a competitive noise floor among reported open-loop structures, while its noise level is slightly higher than that of the closed-loop structure. Fig. 16 shows the measured nonlinearity of 9 fabricated chips, comparing the performance with CCA-only compensation and with the combined CCA and digital calibration. With the CCA enabled, all measured chips achieve a nonlinearity smaller than 3% over the ± 30 g input range even without digital calibration. Variation among chips is mainly due to fabrication-induced mismatch and symmetry errors in the MEMS sensing structure. After enabling the digital calibration, the nonlinearity is further reduced to below 0.3% for all measured samples. These results demonstrate that the proposed combination of circuit-level CCA compensation and lightweight digital calibration can effectively suppress the parasitic-induced nonlinearity and achieve consistent high linearity across multiple chips, making the proposed

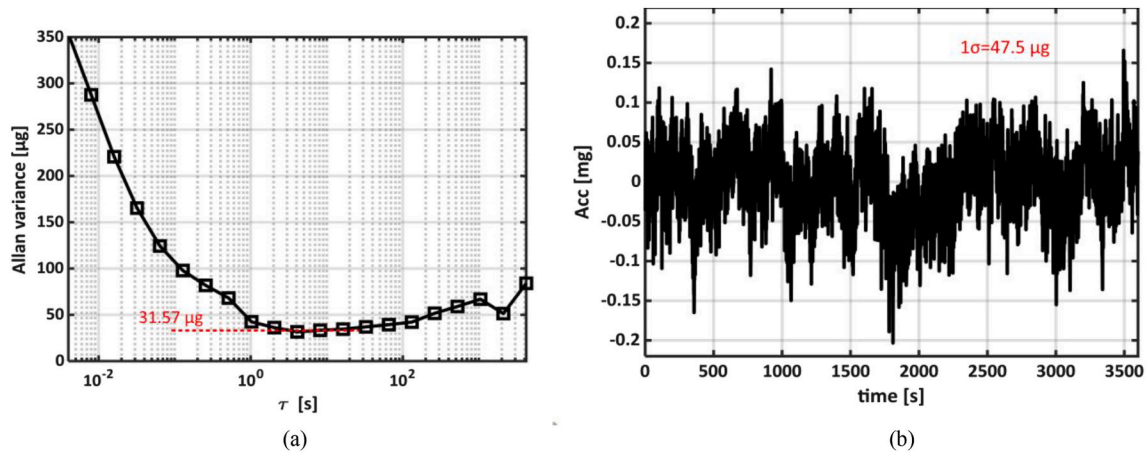
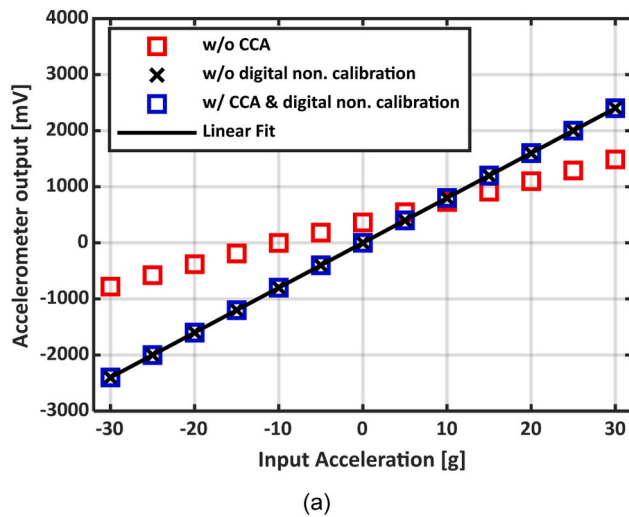
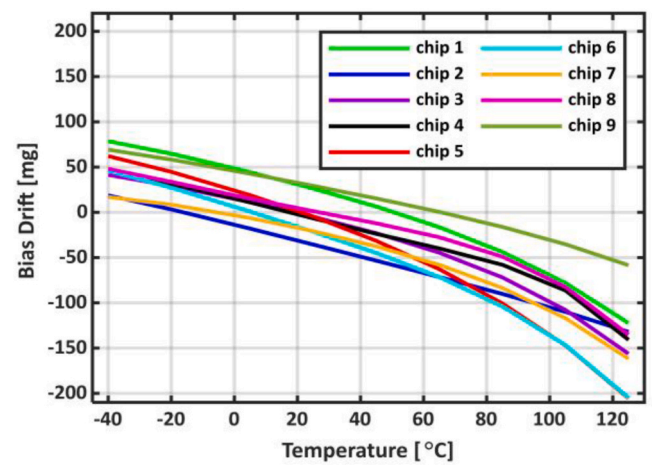


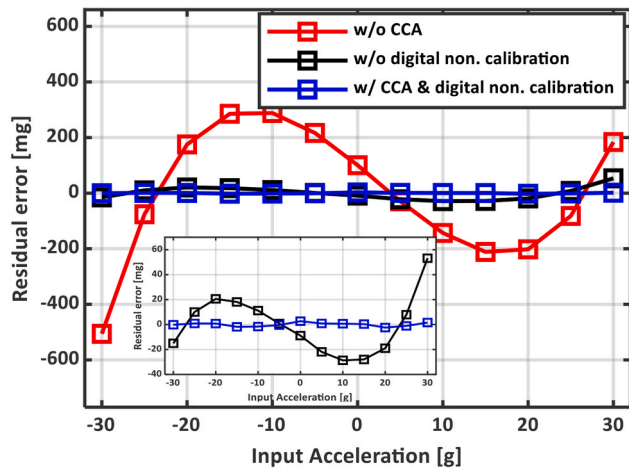
Fig. 13. Measured bias instability of the proposed accelerometer. (a) Allan Deviation analysis result; (b) 1-h zero acceleration output after 1 s average.



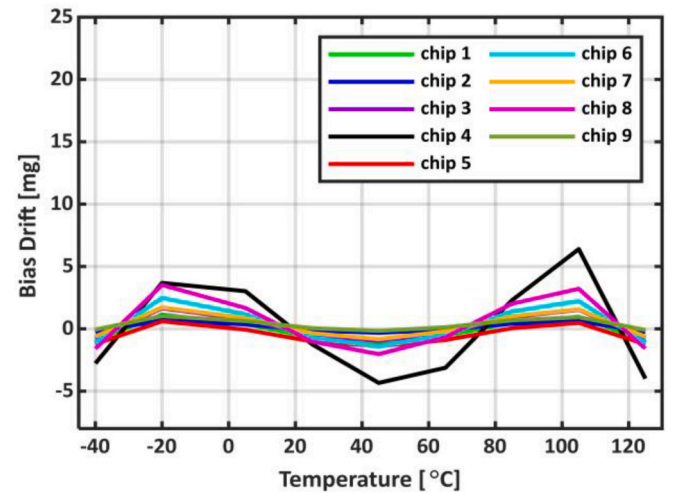
(a)



(a)



(b)



(b)

Fig. 14. DC acceleration sweep measured on a centrifuge: (a) output signal versus input acceleration over a range of ± 30 g with fit curve; (b) residual nonlinearity error under different cases.

Fig. 15. Measured bias drift of the accelerometer over the temperature range of $-40\sim 125$ °C: (a) before calibration (b) after calibration.

design suitable for large-scale manufacturing. In addition, the bias instability performance of this design surpasses that of the closed-loop structure in Ref. [1]. The proposed design also demonstrates superior

power efficiency and FoM, achieving the best FoM among works with a dynamic range exceeding 80 dB. Although [9,10], and [12] report a lower FoM, they target ultra-low-power operation (with low dynamic

Table 1
Performance summary and comparison.

	This work	OJSSC 21 [13]	JSSC 14 [3]	TCASI 22 [9]	CICC 23 [10]	TCASI 23 [12]	TCASI 17 [1]
Structure	Open-loop	Open-loop	Open-loop	Open-loop	Open-loop	Open-loop	Closed-loop
Output Type	Digital	Digital	Digital	Digital	Digital	Digital	Digital
Temperature range [°C]	−40~125	−40~150	−40~135	-	-	−40~80	-
Method	$\Delta C/C$	$\Delta C/C$	$\Delta C/C$	$\Delta C/C$	ΔC	ΔC	ΔC
Circuit Structure	Pseudo-Differential	Single-ended	Fully Differential	Differential	Differential	Differential	Single-ended
Tech [nm]	180	130	180	180	180	180	350
Area [mm ²] ^b	8.32	0.26	1.1	-	5.03	-	12.5
Supply [V]	3.3	2.5	1.9	1.8	1.8	1.8	15/3.3
Core Power [mW]	3.63	1.4	1.39	0.147	0.99	0.036	198.6
Full Scale Range [±g]	30	50	45	12	30	6.5	50
Nonlinearity [%] ^a	1.77/0.089	0.5	3.3	15	-	11	0.623 ^e
Bandwidth [Hz]	200	86	50	2500	8000	1000	200
Noise floor [$\mu\text{g}/\sqrt{\text{Hz}}$]	28.5	84	384	270	57	63	12.7
Dynamic range [dB] ^b	95.47	94.18	82.4	57	73.43	68	106.9
Bias instability [μg]	31.57	-	-	-	8.5	-	200
FoM ₁ [pJ] ^c	305.76	317	2102	83	26.36	14	4471.7
FoM ₂ [$\mu\text{W}\cdot\mu\text{g}/\sqrt{\text{Hz}}$] ^d	7315.4	12,681	75,485	796	630.9	72	178,400

^a Before calibration/After calibration.

^b $\text{DR} = 20 \times \log_{10} (\text{Full scale}/(\text{noise floor} \times \sqrt{(\text{Bandwidth} \times \pi/2)}))$.

^c $\text{FoM}_1 = \text{Power}/(\text{Dynamic range} \times \text{Bandwidth})$.

^d $\text{FoM}_2 = \text{Power} \times \text{noise floor}/\sqrt{\text{Bandwidth}}$.

^e -18 ~ +14 g range.

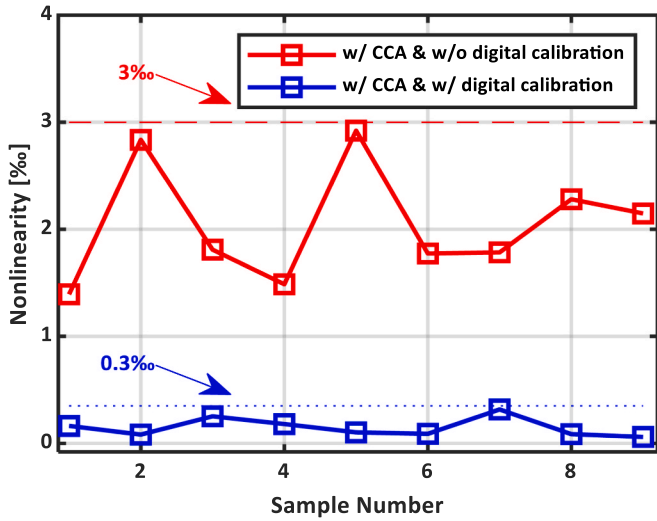


Fig. 16. Measured nonlinearity across 9 fabricated chips: comparison between CCA-only compensation and combined CCA with digital calibration.

range), whereas this work emphasizes a wider dynamic range with a competitive FoM.

5. Conclusion

This work presents a highly integrated readout ASIC for MEMS capacitive accelerometers based on a pseudo-differential four-phase charge-balanced CVC. By exploiting pseudo-differential processing derived from a four-phase clock scheme, the proposed architecture effectively suppresses sensor-originated common-mode interference, bias drift, and sensitivity to supply noise and EMI. To further address the dominant nonlinearity induced by parasitic capacitances, a CCA is incorporated at the front-end C-V conversion stage, providing structural nonlinearity suppression. Combined with a programmable-gain amplifier, a third-order $\Delta\Sigma$ ADC, and on-chip digital temperature and nonlinearity calibration, the proposed system achieves excellent linearity, low noise, and a competitive FoM. These results demonstrate that the proposed architecture effectively bridges the gap between low-cost single-ended sensor interfaces and high-performance readout

solutions, making it well-suited for cost-sensitive and consumer-oriented inertial sensing applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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